

[Menu](#)

Import System

▼ Relevant source files

`import/delimited/importRecordsCSV.js``import/delimited/importRecordsCSV.php``import/utilities/importRecords.js``import/utilities/importRecords.php`

The Heurist Import System enables the ingestion of external data into the Heurist database through structured processes. This document explains the import mechanisms, workflows, and configuration options available for bringing data into Heurist.

Overview

Heurist's import system supports two primary import paths:

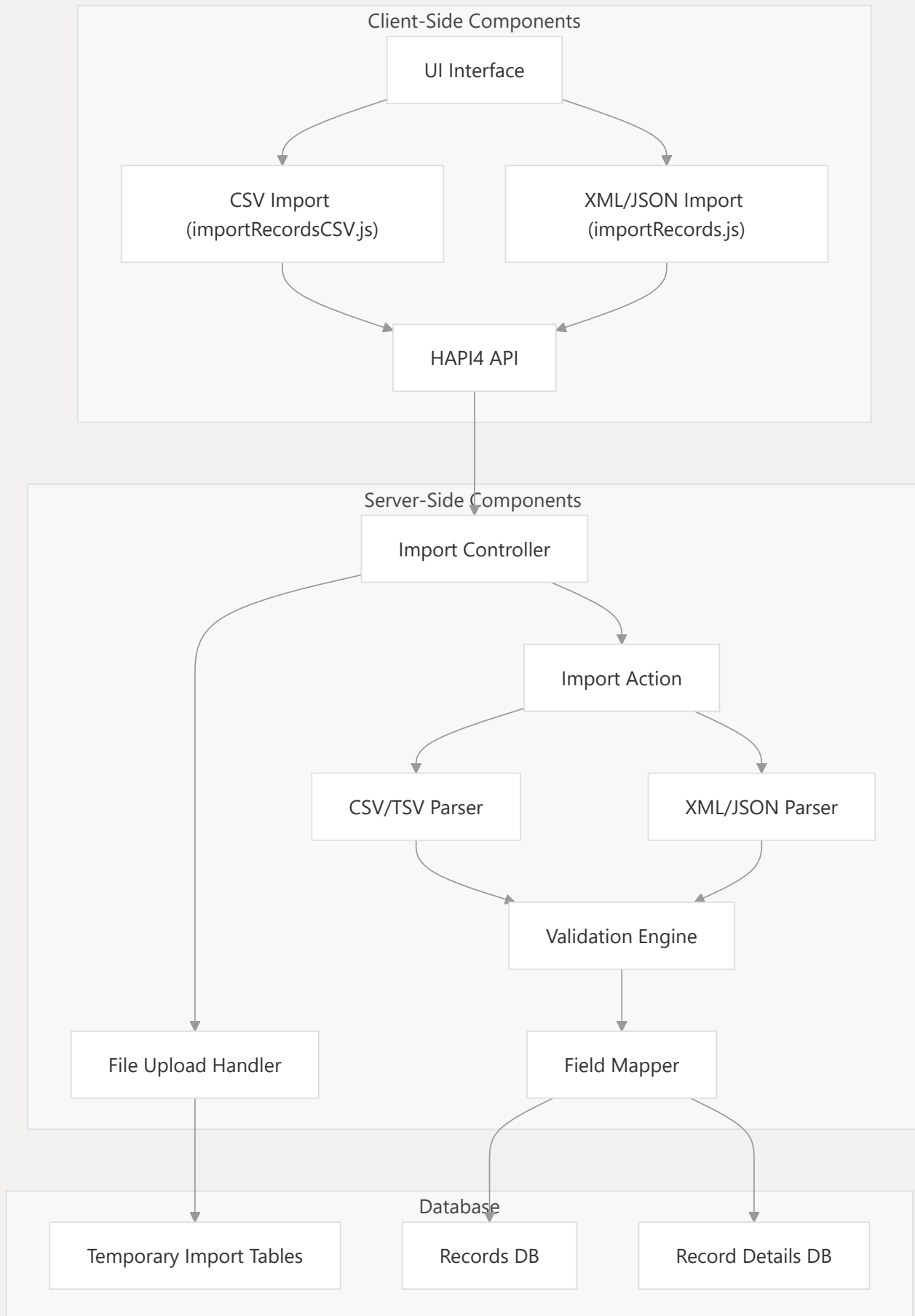
1. **Delimited data import** (CSV/TSV) - For tabular data with each row representing a record
2. **Structured data import** (XML/JSON) - For hierarchical data exported from other Heurist databases or structured sources

The import system includes features for:

- File upload and data validation
- Character encoding detection and conversion
- Field mapping and record type assignment
- Matching against existing records
- Validation of field values against constraints
- Configuration of update behaviors

Sources: `import/delimited/importRecordsCSV.js` 1-56 `import/utilities/importRecords.js` 1-31

Import System Architecture

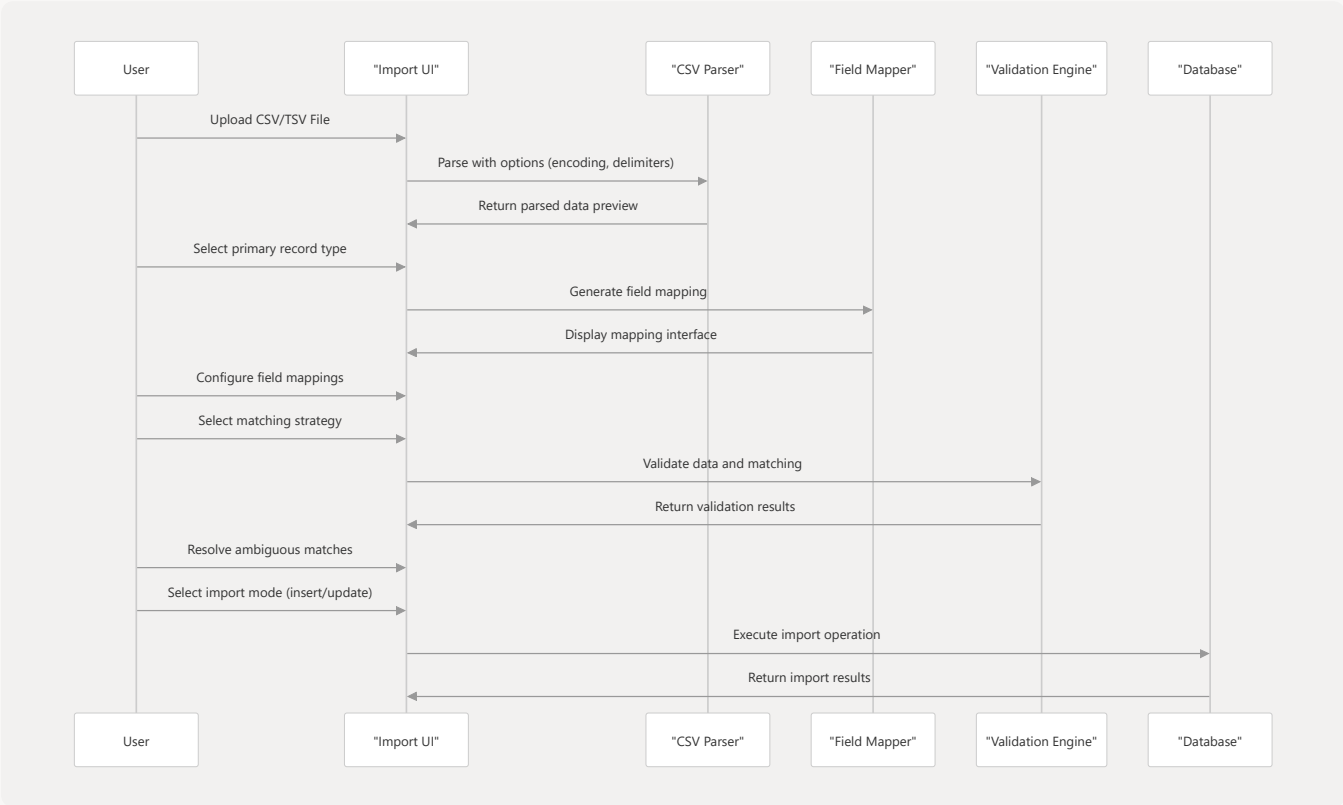


Sources: `import/delimited/importRecordsCSV.js` 23-456

`import/utilities/importRecords.js` 31-100

Delimited Data Import Process (CSV/TSV)

The CSV/TSV import process follows a multi-step workflow that guides users from initial file upload through data validation to final import.



Sources: `import/delimited/importRecordsCSV.js` 23-456

`import/delimited/importRecordsCSV.php` 144-632

Upload and Initial Processing

The CSV/TSV import begins with file upload or direct data entry. The system then processes the data according to user-specified parameters:

- 1. **File Upload:** Supports both file upload and direct data pasting
- 2. **Encoding Detection:** Automatic or manual selection of character encoding
- 3. **Delimiter Configuration:** Options for field separator, text enclosure, line break, and multivalue separator
- 4. **Date Format:** Configurable date format interpretation

Once data is uploaded, the system parses it and presents a preview for user review.

Sources: `import/delimited/importRecordsCSV.js` 65-245

`import/delimited/importRecordsCSV.php` 146-203

Record Type Selection and Field Mapping

After initial parsing, users must:

1. Select the primary record type for the imported data
2. Review dependencies between record types
3. Configure field mapping between CSV columns and Heurist fields

The system can detect potential relationships between record types and generates a sequence for import to respect these dependencies.

Configuration Option	Description
Primary Record Type	Main record type represented by each row of input data
Dependencies	Other record types needed before creating the primary record
Identifier Columns	Columns containing Heurist record IDs (marked with "H-ID")
Date Columns	Columns containing date values requiring special handling

Sources: `import/delimited/importRecordsCSV.js` 456-840

`import/delimited/importRecordsCSV.php` 345-405

Matching and Field Mapping

The matching phase allows the system to identify whether imported data should create new records or update existing ones:

1. **Match on column(s):** Uses column values to match against existing records
2. **Use Heurist ID column:** Directly uses Heurist IDs for matching
3. **Skip matching:** Creates all new records

The field mapping interface allows users to:

- Select which columns to import
- Map columns to specific Heurist fields
- View unique values in each column
- Step through rows to verify data

Sources: `import/delimited/importRecordsCSV.js` 320-455

`import/delimited/importRecordsCSV.php` 467-479

Import Configuration and Execution

Before finalizing the import, users can:

1. Validate the data against Heurist constraints
2. Resolve errors and warnings
3. Select the import mode:
 - Add new values (ignoring duplicates)
 - Replace all existing values
 - Add new values only to empty fields
 - Replace values with retention options

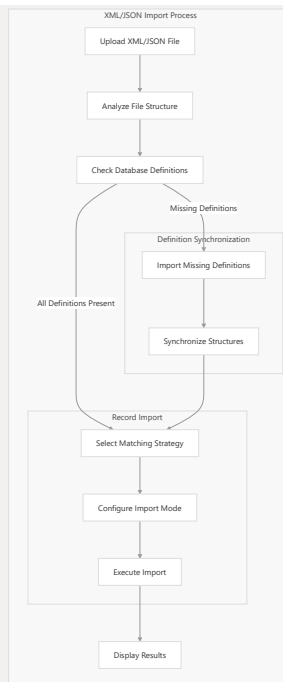
The system reports on the number of records to be inserted, updated, or ignored.

Sources: `import/delimited/importRecordsCSV.js` 419-435

`import/delimited/importRecordsCSV.php` 497-546

Structured Data Import (XML/JSON)

The XML/JSON import process is designed for importing data from other Heurist databases or structured data sources.



Sources: `import/utilities/importRecords.js` 31-230 `import/utilities/importRecords.php` 86-266

File Analysis and Definition Synchronization

The system analyzes the uploaded file to extract:

1. Source database information (ID, name, URL)
2. Record types and counts in the file
3. Field types used in the file

The system then checks if all required definitions (record types and field types) exist in the target database. If missing definitions are found, users can:

1. Download the definitions from the source database
2. Proceed with import (ignoring data for missing definitions)
3. Download a list of elements in the source file for manual verification

Sources: `import/utilities/importRecords.js` 148-179 `import/utilities/importRecords.js` 230-374

Record Import Configuration

Once definitions are synchronized or verified, users can configure the import:

1. **Identifier Field:** Select a field to use for identifying duplicate records
2. **Import Mode:**
 - Create new records

- Update existing records
- Both create and update

For updates, users can specify how existing data should be handled:

Update Mode	Description
Add new values without deleting existing	New values added as repeats, duplicates ignored
Replace all existing values	Completely replace existing field values
Add only if field is empty	Only populate empty fields
Replace with retention	Replace values but keep existing if no new value provided

Sources: `import/utilities/importRecords.js` 539-640

`import/utilities/importRecords.php` 212-233

Technical Implementation

Import Session Management

The import system maintains import sessions to allow users to return to in-progress imports. Each import is assigned a session ID (`imp_ID`), and session information is stored in the `sysImportFiles` entity.

Key session information includes:

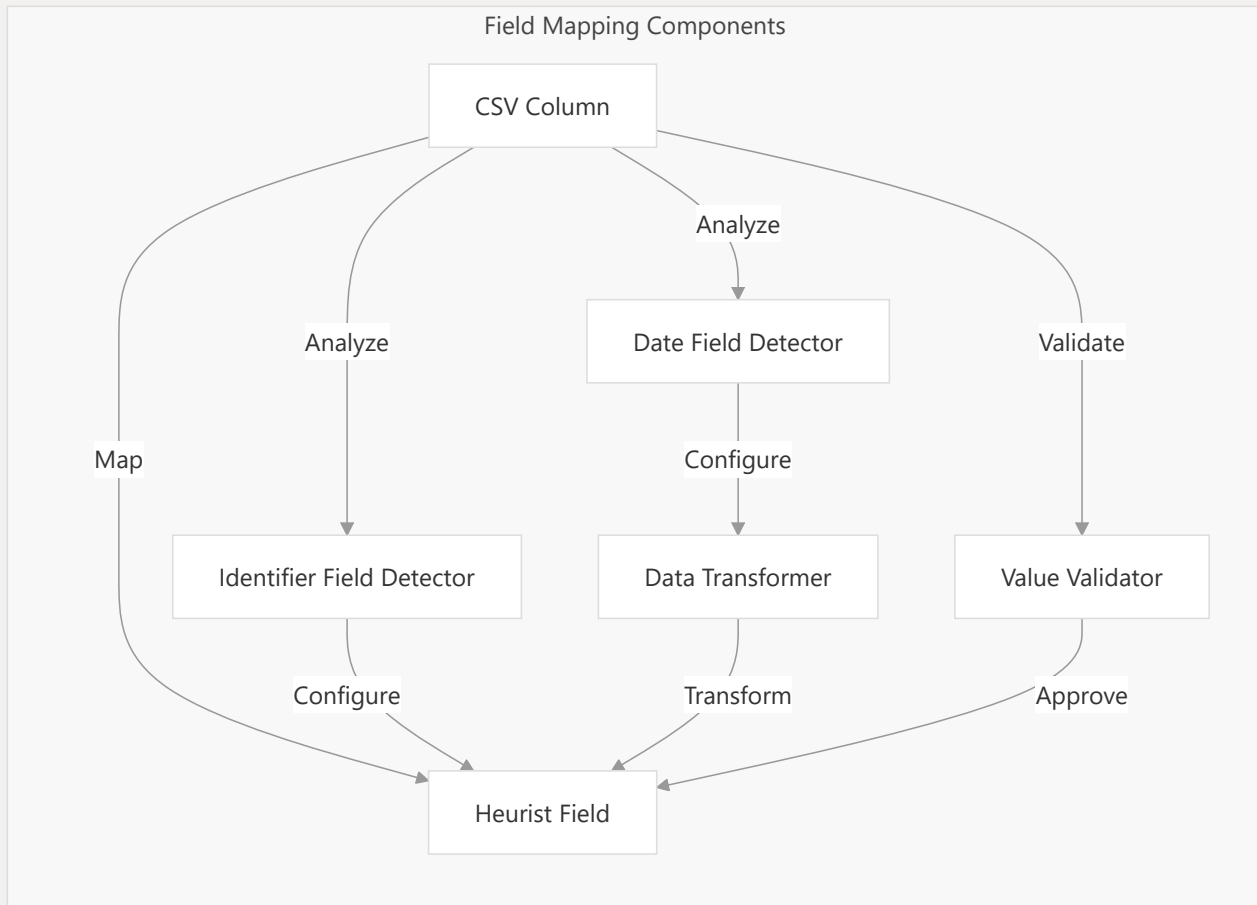
- Original file name
- Processing information (JSON-encoded)
- Import configuration (field mappings, record types)
- Sequence of record types to be imported

Sources: `import/delimited/importRecordsCSV.js` 27-45

`import/delimited/importRecordsCSV.js` 463-555

Field Mapping and Validation

The field mapping component is central to the import process:



Field mapping includes:

1. Detection of special field types (IDs, dates)
2. Validation of values against constraints
3. Transformation of data to match Heurist requirements
4. Configuration of multi-value handling

Sources: `import/delimited/importRecordsCSV.js` 840-950

`import/delimited/importRecordsCSV.php` 309-328

Record Type Dependencies

The import system handles complex dependencies between record types:

1. **Dependency Analysis:** Identifies which record types depend on others through pointer fields
2. **Import Sequence:** Creates a sequence to ensure dependent records are imported first
3. **Hierarchical Relationships:** Supports multi-level dependencies

The system renders these dependencies as a sequence ribbon in the UI, allowing users to visualize and navigate through the import process.

Sources: `import/delimited/importRecordsCSV.js` 734-840

`import/delimited/importRecordsCSV.js` 918-1002

Error Handling and Validation

The import system provides comprehensive error handling:

1. **Warnings:** Issues that don't block import but may indicate problems
2. **Errors:** Issues that must be resolved before import